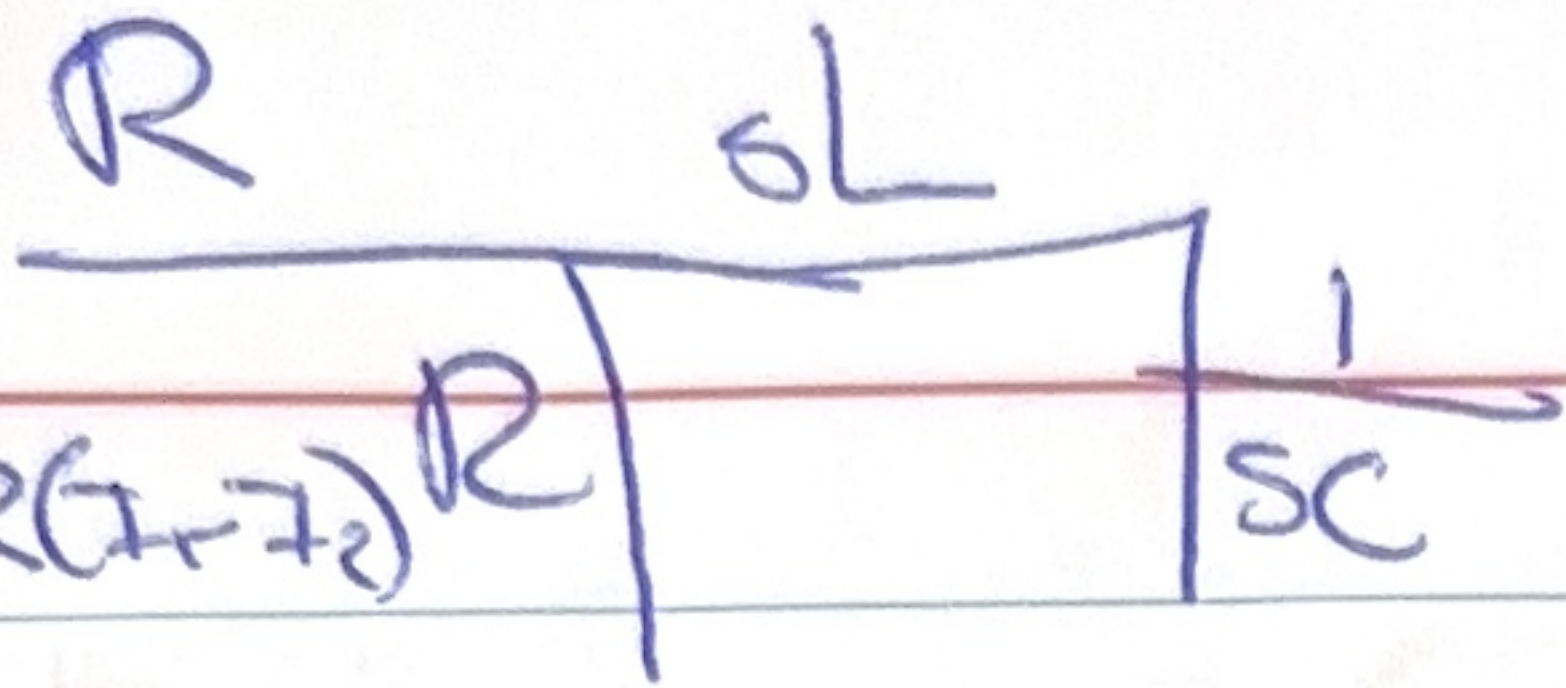
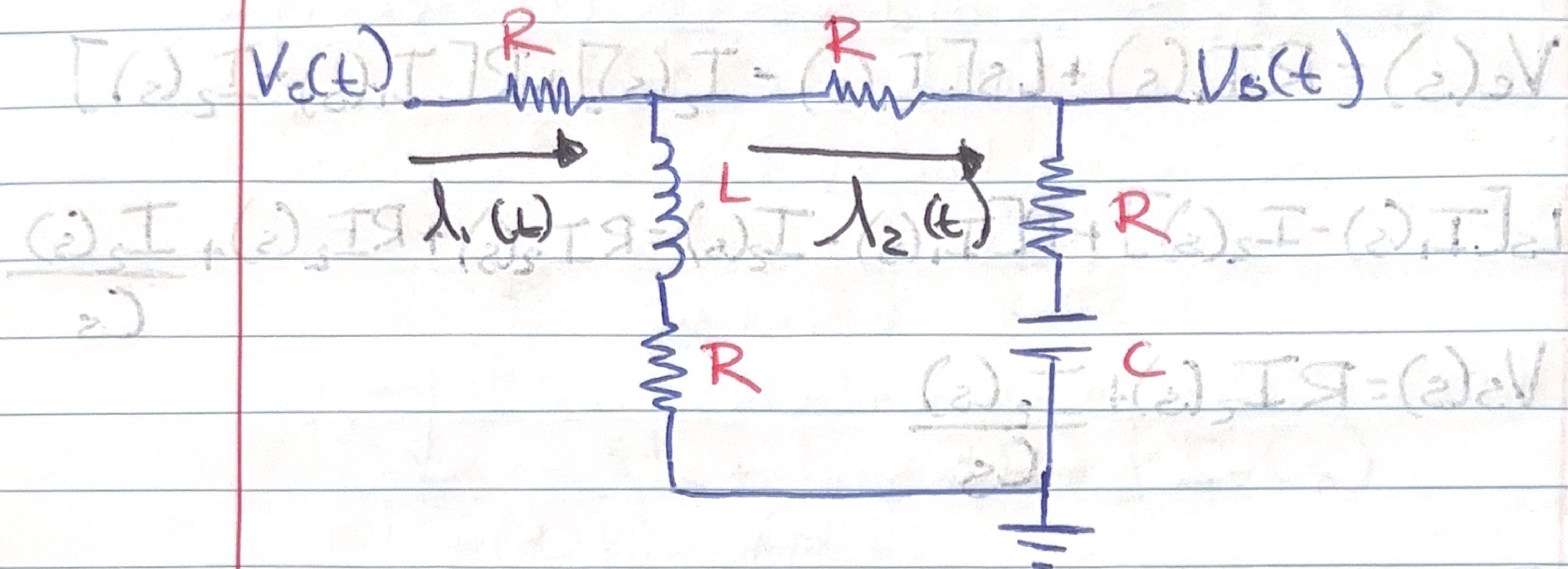




$$V_c = V_K + I R (T_1 - T_2)$$



Ecuaciones principales.

$$V_c(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) + i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) +$$

$$\frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = RI_1(s) + Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)]$$

$$Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = RI_2(s) + RI_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = RI_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico

Función de transferencia

- Agrupar corrientes $I_1(s)$ e $I_2(s)$
- Expresar el $V_e(s)$ en función de $I_2(s)$
- Despejar $I_1(s)$ de la segunda ecuación principal

$$V_s(s) = \left(R + \frac{1}{Cs}\right) I_2(s) = \left(\frac{CRs + 1}{Cs}\right) I_2(s)$$

$$LsI_1(s) - LsI_2(s) + RI_1(s) - RI_2(s) = RI_2(s) + \frac{I_2(s)}{Cs}$$

$$LsI_1(s) + RI_1(s) = RI_2(s) + RI_2(s) + \frac{I_2(s)}{Cs} + LsI_2(s) + RI_2(s)$$

$$(Ls + R)I_1(s) = \left(Ls + \frac{1}{Cs} + 3R\right)I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R \right) I_2(s) \quad (1)$$

$$I_1(s) \left(\frac{Ls}{Ls + R} + \frac{1}{Cs(Ls + R)} + \frac{3R}{Ls + R} \right) I_2(s)$$

$$I_1(s) = \left[\frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R \right) \right] I_2(s)$$

$$I(s) = \frac{(Ls^2 + 3CRs + 1)}{CLs^2 + CRs}$$

$$V_e(s) = RI_1(s) + LsI_1(s) - LsI_2(s) + RI_1(s) - RI_2(s)$$

$$V_e(s) = (Ls + 2R)I_1(s) - (Ls + R)I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{(Ls^2 + 3CRs + 1)}{CLs^2 + CRs} I_2(s) - (Ls + R)I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1)}{CLs^2 + CRs} - (Ls + R) \right] I_2(s)$$

$$2R \cdot 3CRs$$

$$6CR^2s$$

$$V_c(s) \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - (Ls + R)(CLs^2 + CRs)}{CL^2s + CRs} \right] I_2(s)$$

$$= \frac{(Ls^3 + 3CLR s^2 + Ls + 2RLs^2 + 6CR^2s - (Ls^3 - CLR s^2 - CLR s^2 - CR^2s))}{CL^2s + CRs}$$

$$= \left[\frac{Ls^3 + CLR s^2 + Ls + 2RLs^2 + 5CR^2s - (Ls^3)}{CL^2s + CRs} \right] I_2(s)$$

$$= \left[\frac{(Ls^2 + CLR s + L + 2RLs + 5CR^2 - (Ls^2))s}{CL^2s + CRs} \right] I_2(s)$$

$$3CLR s^2 + (L + 5CR^2)s + 2R$$

$$(2) V_c(s) = \left[\frac{3CLR s^2 + (L + 5CR^2)s + 2R}{CLs^2 + CRs} \right] I_2(s)$$

$$(2) I(s+2) - (2) I(s+2) = (2) V$$

$$V_s(s) = \frac{(Rs + 1)}{CLs^2 + CRs} I_2(s)$$

$$(2) V_c(s) = \frac{3CLR s^2 + (L + 5CR^2)s + 2R}{CLs^2 + CRs} I_2(s)$$

$$(2) I \left[\frac{(s+2)(1+e^{sT})(s+2)}{(s+2)(1+e^{sT})(s+2)} \right] = (2) V$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(CRs+1)(Ls+R)}{3CLR s^2 + (L+5CR^2)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR s^2 + CR^2 s + Ls + R}{3CLR s^2 + (L+5CR^2)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR s^2 + (L+CR^2)s + R}{3CLR s^2 + (L+5CR^2)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(CRs+1)I_2(s)}{3CLR s^2 + (5CR^2+L)s + 2R} \cdot \frac{Ls+R}{Ls^2+CRs}$$

For a physical or basic circuit, the transfer function is a ratio of polynomials in s .

$$R = 10 \Omega$$

$$L = 0.5 \text{ mH}$$

$$C = 10^{-4} \text{ F}$$

$$3CLR s^2 + (5CR^2+L)s + 2R = 0$$

$$b^2 - 4ac$$

$$\times 10^{-6})(10 \times 10^3)$$

$$0^3)^2 + 220 \times 10^{-6}$$

$$235000.00002$$

$$7 \rightarrow -75.75757567 \times 10^6$$

Estado estacionario

Error en estado estacionario cuando se aplica un escalón unitario de entrada.

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R} \right) = \frac{R}{2R}$$

$$e(t) = 0.5V$$

Estabilidad en lazo abierto
Se calculan los polos del sistema (raíces del denominador)

$$R = 10k\Omega$$

$$L = 220\mu H$$

$$C = 470\mu F$$

$$3CLR s^2 + (5CR^2 + L)s + 2R = 0$$

$$X_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2a

$$a = 3CLR$$

$$b = 5CR^2 + L$$

$$c = 2R$$

$$a = 3(470 \times 10^{-6})(220 \times 10^{-6})(10 \times 10^3)$$

$$a = 3.102 \times 10^{-3}$$

$$b = 5(470 \times 10^{-6})(10 \times 10^3)^2 + 220 \times 10^{-6}$$

$$b = 0.235 \times 10^6 \rightarrow 235000.00002$$

$$c = 2(10 \times 10^3)$$

$$c = 20 \times 10^3$$

$$S_1 = -0.08510$$

$$S_2 = -75757575.67 \rightarrow -75.75757567 \times 10^6$$

$$X = -$$

